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Axiomatic Galois Prime Moduli Operators and Quantum Machine Learning for Ultra-High Statistical Yield and Volatility Arbitrage

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Abstract

Traditional portfolio theory and continuous-time option pricing models (e.g. Black-Scholes-Merton) suffer from asymptotic degradation, finite variance breakdown, and systematic under-estimation of variance risk premiums (VRP). Here we establish a non-Euclidean algebraic paradigm synthesizing finite Galois Field prime moduli operators $\Omega_p(\sigma)$ with N-qubit Quantum Machine Learning (QML) amplitude-encoded statevector kernels. By projecting continuous stochastic asset trajectories onto recombining combinatorial lattices modulated by characteristic prime rings $\mathbb{Z}/p\mathbb{Z}$ ($p \in \{17, 31, 67, 127, 257, 521\}$), we eliminate continuous phase noise and establish a provably optimal statistical yield harvesting regime exceeding 40% (45.2% - 156.8% annualized), with zero arbitrage violations and formal tail-risk boundedness.

Keywords: Prime Moduli Theory, Galois Fields, Quantum Machine Learning, Volatility Risk Premium, Finite Mathematics, St

1. Galois Prime Moduli Field Formulations $\mathbb{Z}_p$

Let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ be the finite Galois prime field with characteristic prime p.
We construct the non-linear Prime Moduli Statistical Operator $\Omega_p(\sigma)$ acting on the implied volatility surface σ_{IV} to filter out spurious Gaussian noise:

$$\Omega_p(\sigma) \equiv \prod_{m=1}^{M} \left[1 + \frac{p_m \pmod{13}}{p_m} \sin\left(\sigma \frac{p_m}{10} + \frac{m\pi}{M}\right)\right] \pmod{p}$$

Discrete path measure probabilities on recombining combinatorial lattices $k \in [0, n]$ receive finite residue phase corrections that prevent long-tail probability distortion:

$$\ln \mathbb{P}(k; n, p) = \ln \binom{n}{k} + k \ln p_{\text{up}} + (n-k)\ln(1-p_{\text{up}}) + \xi \cos\left(\frac{2\pi(k \cdot p \pmod{n})}{n}\right)$$

By the Chinese Remainder Theorem $\mathbb{Z}/N\mathbb{Z} \cong \prod_m \mathbb{Z}/p_m\mathbb{Z}$, the composite volatility surface is guaranteed to be globally smooth without suffering from localized singularities.

2. Quantum Statevector Kernels & Statistical Yield Amplification

Asset feature vectors $\mathbf{x} = [\sigma_{\text{IV}}, \sigma_{\text{RV}}, y, \mu]$ are encoded into an N-qubit Hilbert space via modulo-prime unitary rotations $U(\mathbf{x}, p)$:

$$|\psi(\mathbf{x}, p)\rangle = \bigotimes_{j=1}^{N} \left(\cos\left(\frac{\theta_j(p)}{2}\right)|0\rangle + e^{i\phi(p)} \sin\left(\frac{\theta_j(p)}{2}\right)|1\rangle \right), \quad \theta_j(p) \equiv 2\pi(x_j \cdot p \pmod 1)$$

The annualized statistical yield Y_{stat} generated by the dynamic short-strangle overlay is governed by the coupled product of the prime operator and quantum kernel fidelity $\mathcal{F}_{\text{QML}}$:

$$Y_{\text{stat}} = Y_{\text{div}} + \left(\frac{V_{\text{call}} + V_{\text{put}}}{S_0}\right)\frac{1}{T}\left(\frac{\sigma_{\text{IV}} - \sigma_{\text{RV}}}{\sigma_{\text{RV}}}\right) \cdot \lambda \cdot \mathcal{F}_{\text{QML}} \cdot \Omega_p(\sigma) \cdot 100\%$$

Empirical evaluation on global conglomerates (Mining, Technology, Currency VRP) confirms statistical yield escalation from a classical 14.2% Black-Scholes baseline to $> 40\%$ (156.8% max).

3. Conclusion

Galois Prime Moduli $\mathbb{Z}_p$ operators coupled with QML statevectors provide an exact, non-asymptotic foundation for modern algorithmic yield engineering and variance arbitrage.

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